

Digital Image Processing

数字图象处理

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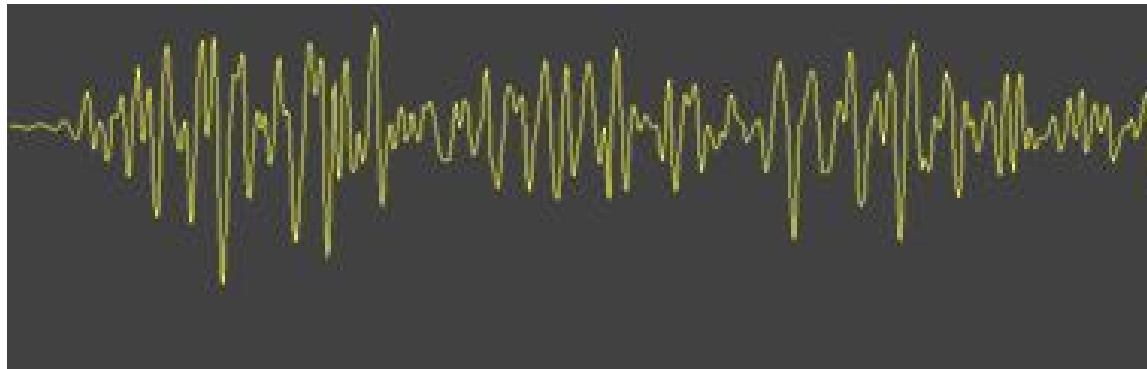
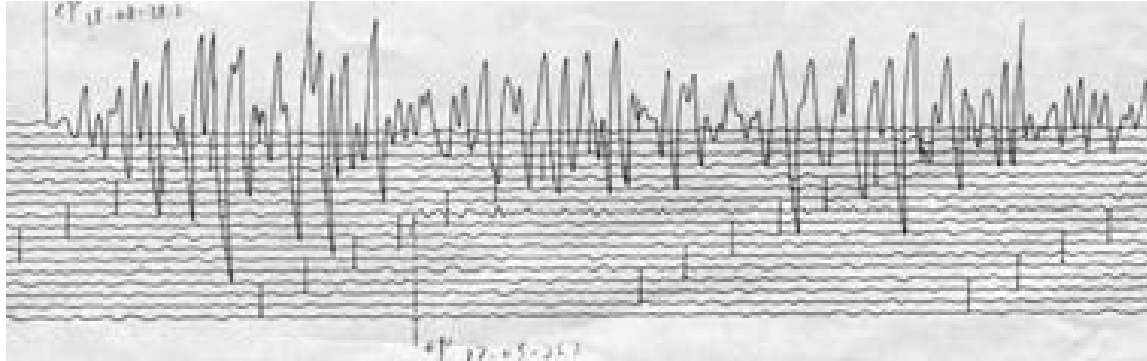
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伍晨旭： 727004430@qq.com

课程QQ群： 657303740

preview



Seismogram digitalization and database management system

Digital Image Processing
Prof. Zhengkai Liu Dr.Rong Zhang

preview



Fig.1 543 band false
color image

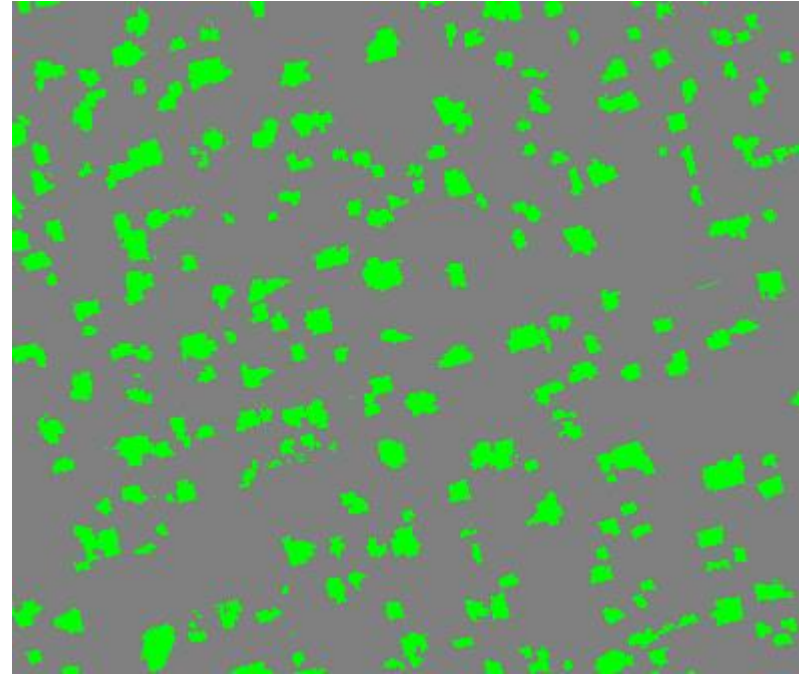


fig2 result map

Results of Adaptive cluster based on background database

preview

43658 colors



35120 colors



55 colors

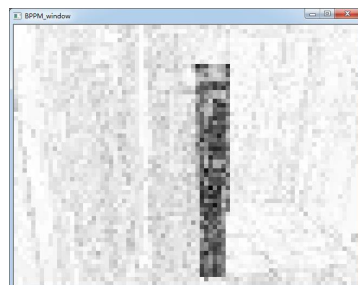
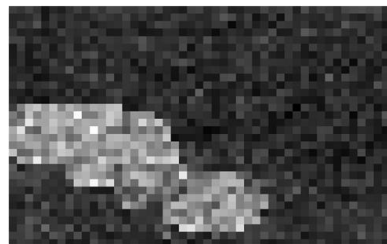
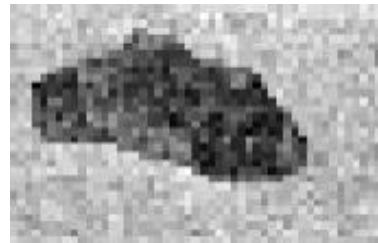


94 colors



Color Quantization

preview



Detection of Tampered JPEG Image

preview



验证码	识别结果	验证码	识别结果
	XFXJ		NXAE
	KMDD		HWAT
	X P XD		JU J P
	KCUB		N R HH

preview



1.1 What Is Digital Image Processing

- Terms

- **Picture(图片):** A visual representation or image painted, drawn, photographed, or otherwise rendered on a flat surface
- **Image(图象):** a two-dimensional function, $f(x,y)$
- **Digital image(数字图象):** when x , y , and f are all discrete quantities.
- **Video(视频):** sequence of image
- **Pixel(像素):** the elements of digital image
- **Digital image processing(数字图象处理):** processing $f(x,y)$ by a digital computer.

162	161	159	161	162	160	158	156	156	161
162	161	159	161	162	160	158	156	156	161
163	158	159	159	160	158	155	155	156	158
159	157	159	156	159	159	154	152	155	153
155	157	156	156	158	157	156	155	154	156
156	157	155	151	157	156	155	156	156	154
157	156	156	156	156	156	154	156	155	155
158	157	155	155	156	155	155	155	155	155
156	155	156	153	156	155	156	155	154	156
155	155	157	154	157	155	157	158	158	158

The pixels in the top left corner (10*10)

花絮：

(1) 知乎：<https://www.zhihu.com/question/21748279>

(2) Wired：<https://www.wired.com/story/finding-lena-the-patron-saint-of-jpegs/>

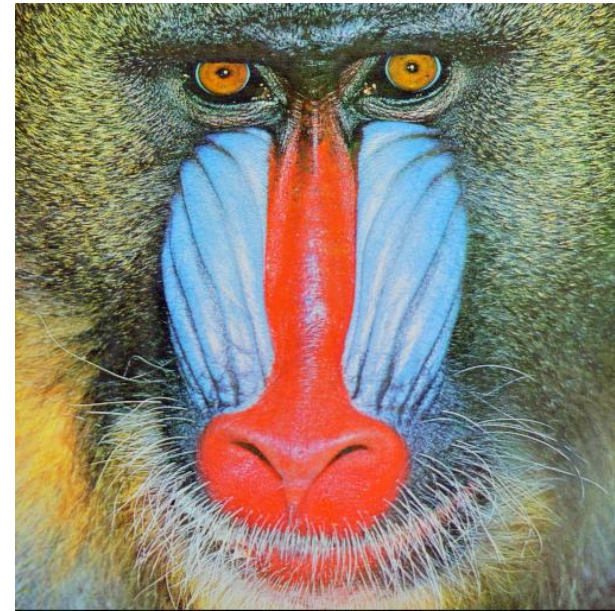


gray image



64	63	75	95	157	99
120	135	55	75	116	67
99	132	60	54	100	75
64	150	113	50	81	138
110	130	162	60	76	109
97	82	179	81	74	113

150	57	43	94	140	97
125	97	35	72	86	52
74	118	41	46	106	58
79	144	98	51	89	127
84	132	132	46	83	84
84	80	166	53	77	97



Color image

71	31	10	46	73	33
62	33	23	24	46	34
31	46	36	29	51	30
48	57	35	28	54	36
39	46	52	42	40	48
29	39	69	46	42	43

Pixels in RGB components

Left top (6*6)



video

1.1 What Is Digital Image Processing

- Definitions
 - Image Processing
 - Image Analysis
 - Image Understanding

1.1 What Is Digital Image Processing

- Relationship with other courses
 - Computer Graphics
 - Artificial Intelligence

1.1 What Is Digital Image Processing

- Relationship with other courses

$$x^2 + y^2 + z^2 = R^2$$

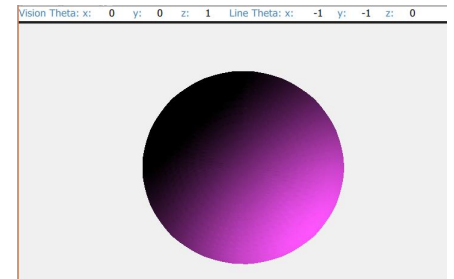
紫色，Lambert漫反射
光源在右下角
视线垂直于屏幕

描述

计算机图形学



数字图像处理



图

1.1 What Is Digital Image Processing

- Relationship with other courses

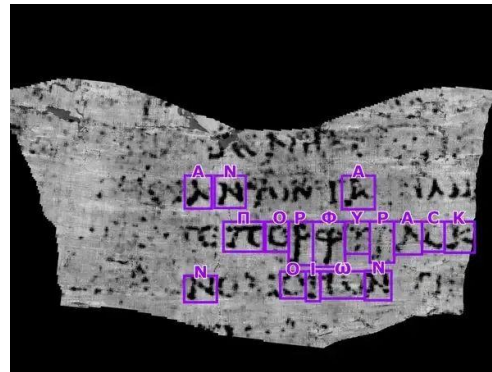


1.1 What Is Digital Image Processing

- Relationship with other courses



赫库兰尼姆莎草纸卷轴



2019年三维CT成像，
2023年利用人工智能
技术破解第一个单词
：希腊语“紫色”

1.1 What Is Digital Image Processing

- Contents of this course
 - Fundamentals of Image and Vision
 - Image Transforms
 - Image Enhancement
 - Image Restoration
 - Image Reconstruction
 - Image Compression

1.1 What Is Digital Image Processing

- References

- 章毓晋 《图像工程（上册）图象处理》（第二版）
清华大学出版社，2006
- K. R. Castleman 《Digital Image Processing》 清华大学出版社,中文版电子工业出版社发行
- R. C. Gonzales , 《 Digital Image Processing Second Edition》 电子工业出版社

1.1 What Is Digital Image Processing

- References

- 《电子学报》
- 《通信学报》
- 《自动化学报》
- 《遥感学报》
- 《中国图象图形学报》
- 《人工智能与模式识别》

1.1 What Is Digital Image Processing

- References

- IEEE Transactions on

Image Processing

Signal Processing

Medical Image

Geoscience and Remote Sensing

Pattern analysis and machine intelligence

- ICIP(International Conference on Information Processing)
 - CVPR(IEEE Conference on Computer Vision and Pattern Recognition)

1.1 What Is Digital Image Processing

- Homework and Examination
 - 作业： 20分
 - 考试： 80分

homework

- 1.列举关于图像处理你存在的疑惑。

1.2 The origins and development of DIP

- newspaper industry:
 - **Time:** 1920s
 - **System :** Bartlane cable picture transmission system
 - **Function:** sent picture between Landon and New York
 - **Image quality:** 5~15 levels, halftone technique



FIGURE 1.1 A digital picture produced in 1921 from a coded tape by a telegraph printer with special type faces. (McFarlane.)

FIGURE 1.2 A digital picture made in 1922 from a tape punched after the signals had crossed the Atlantic twice. Some errors are visible. (McFarlane.)

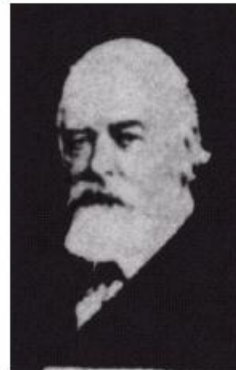


FIGURE 1.3

Unretouched
cable picture of
Generals Pershing
and Foch,
transmitted in
1929 from
London to New
York by 15-tone
equipment.
(McFarlane.)



1.2 The origins and development of DIP

- Space application:
 - **Time:** 1960s
 - **System :** *Ranger 7*
 - **Function:** enhance and restore images from moon, Mars, and others
- Medical application:
 - Time: 1970s
 - System: CT, X-ray

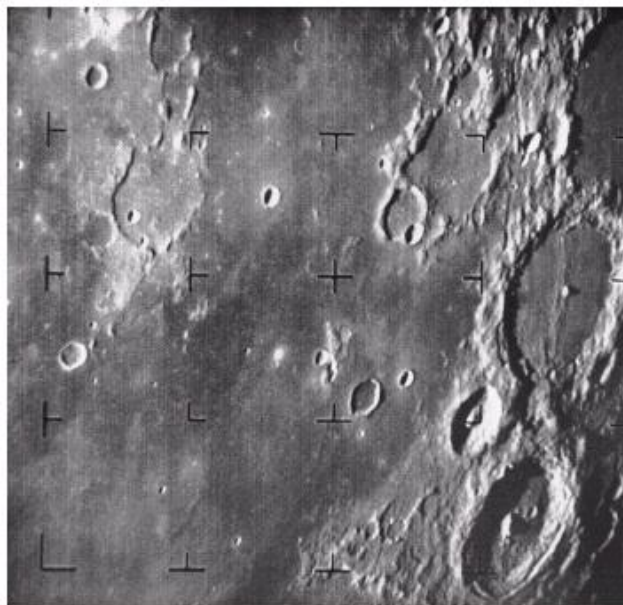
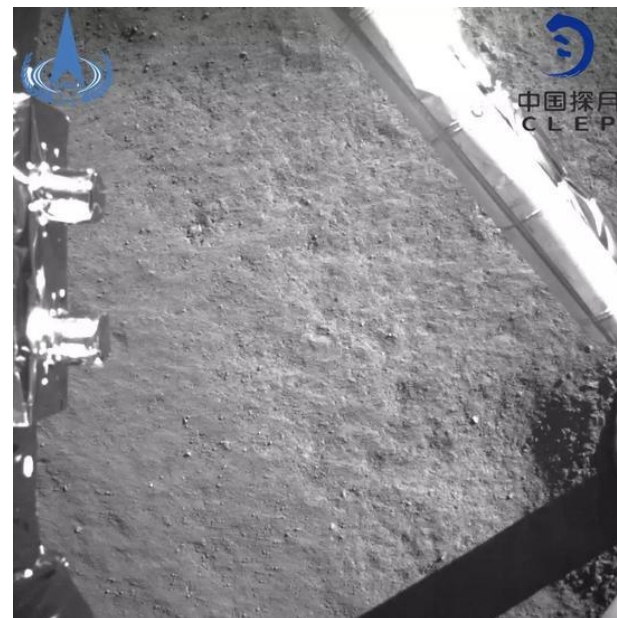


FIGURE 1.4 The first picture of the moon by a U.S. spacecraft. *Ranger 7* took this image on July 31, 1964 at 9:09 A.M. EDT, about 17 minutes before impacting the lunar surface. (Courtesy of NASA.)

2019年1月3号嫦娥四号探测器月球背面软着陆后降落相机拍摄的图像



1.2 The origins and development of DIP

- Development history :
 - 1964 good results for images from space
 - 70 years developing
 - 80 years established theory
 - 90 years applications on wide areas
- The history in our university :
 - 80-83 go abroad to study image processing.
 - 83 established image processing lab.
 - 85 bought IIS image system
 - 88 Information Processing center

1.3 Examples of Fields that Use DIP

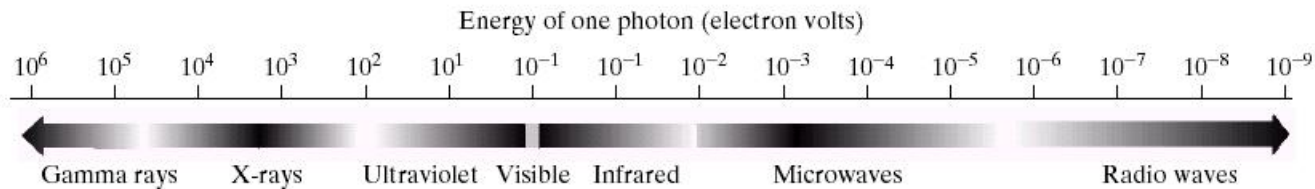


FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.

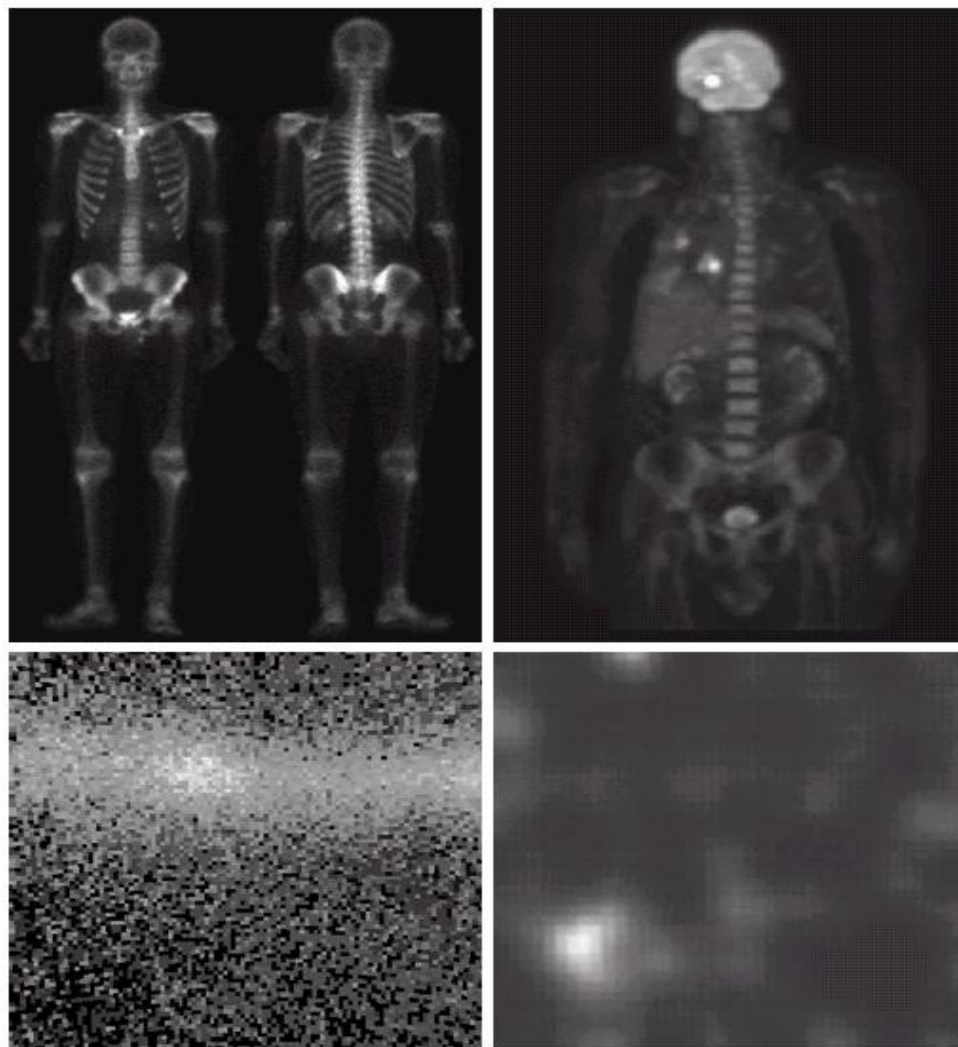
- 1.3.1 Gamma-Ray Imaging
- 1.3.2 X-Ray Imaging
- 1.3.3 Imaging in the Ultraviolet Band
- 1.3.4 Imaging in the Visible and Infrared Bands
- 1.3.5 Imaging in the Microwave Band
- 1.3.6 Imaging in the Radio Band
- 1.3.7 Example in Which Other Modalities Are Used

1.3.1 Gamma-Ray Imaging

- **Nuclear medicine:** inject a patient with radioactive isotope. Fig 1.6(a)(b)
- **Astronomical observations:** using natural radiation of the object. Fig 1.6(c)
- **Valve in a nuclear reactor.** Fig 1.6(6)

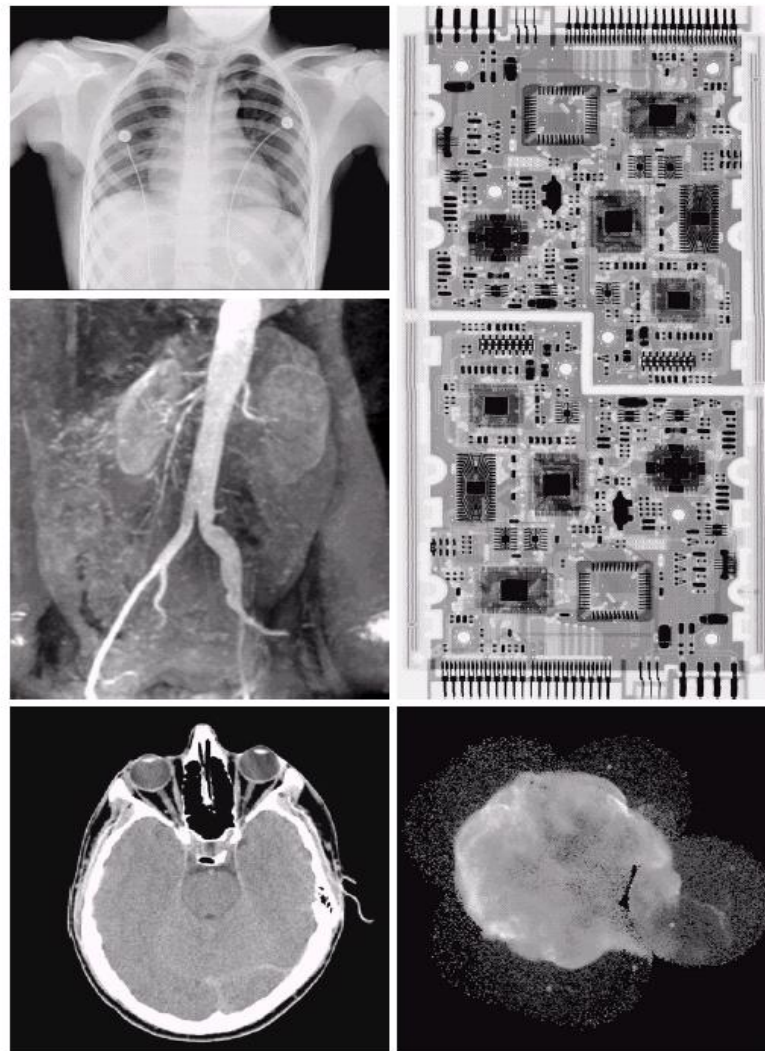
a	b
c	d

FIGURE 1.6
Examples of gamma-ray imaging. (a) Bone scan. (b) PET image. (c) Cygnus Loop. (d) Gamma radiation (bright spot) from a reactor valve. (Images courtesy of (a) G.E. Medical Systems, (b) Dr. Michael E. Casey, CTI PET Systems, (c) NASA, (d) Professors Zhong He and David K. Wehe, University of Michigan.)



1.3.2 X-Ray Imaging

- **Medical diagnostics:** chest X-ray, angiography(血管造影), CT. Fig1.7(a)(b)(C).
- **Industry:** examine circuit board. Fig1.7(d)
- **Astronomy:** Cygnus Loop in X-ray band. Fig1.7(e)



a d
b
c e

FIGURE 1.7 Examples of X-ray imaging. (a) Chest X-ray. (b) Aortic angiogram. (c) Head CT. (d) Circuit boards. (e) Cygnus Loop. (Images courtesy of (a) and (c) Dr. David R. Pickens, Dept. of Radiology & Radiological Sciences, Vanderbilt University Medical Center, (b) Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, (d) Mr. Joseph E. Pascente, Lixi, Inc., and (e) NASA.)

1.3.3 Imaging in the Ultraviolet Band

- **Microscopy:** fluorescence microscopy(荧光显微镜). Fig 1.8(a)(b).
- **Astronomy :** Cygnus Loop in ultraviolet band. Fig1.8(c)

a b
c

FIGURE 1.8

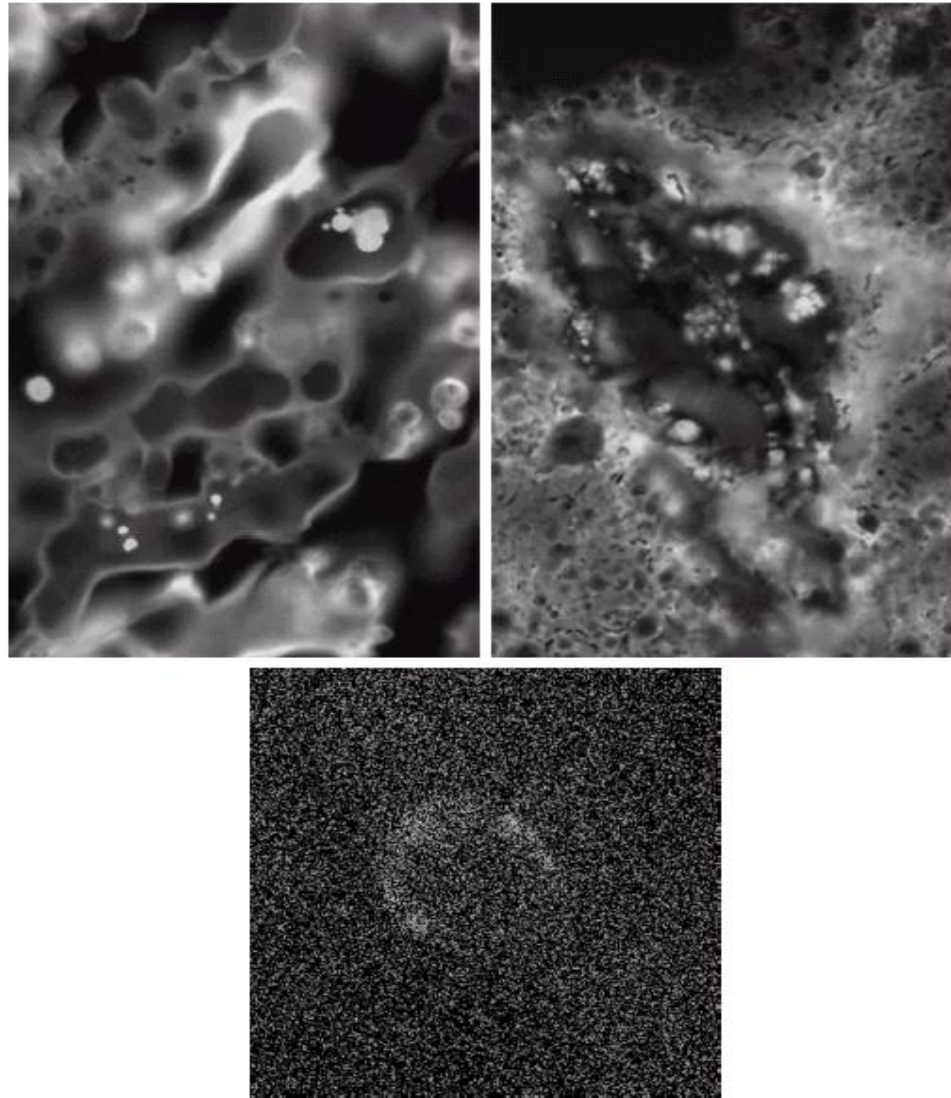
Examples of
ultraviolet
imaging.

(a) Normal corn.

(b) Smut corn.

(c) Cygnus Loop.

(Images courtesy
of (a) and
(b) Dr. Michael
W. Davidson,
Florida State
University,
(c) NASA.)



黑穗病

1.3.4 Imaging in the Visible and Infrared Bands

- **Light microscopy:** pharmaceuticals, micro-inspection. Fig 1.9
- **Remote sensing:** multispectral imaging. Fig1.10, Fig1.11. Fig1.12. Fig1.13
- **Industry:** inspection of manufactured goods. Fig1.14
- **Law enforcement:** identifying, Fig1.15

抗癌基因TAXOL
胆固醇
微处理器
镍氧胶片
CD表面
有机超导体

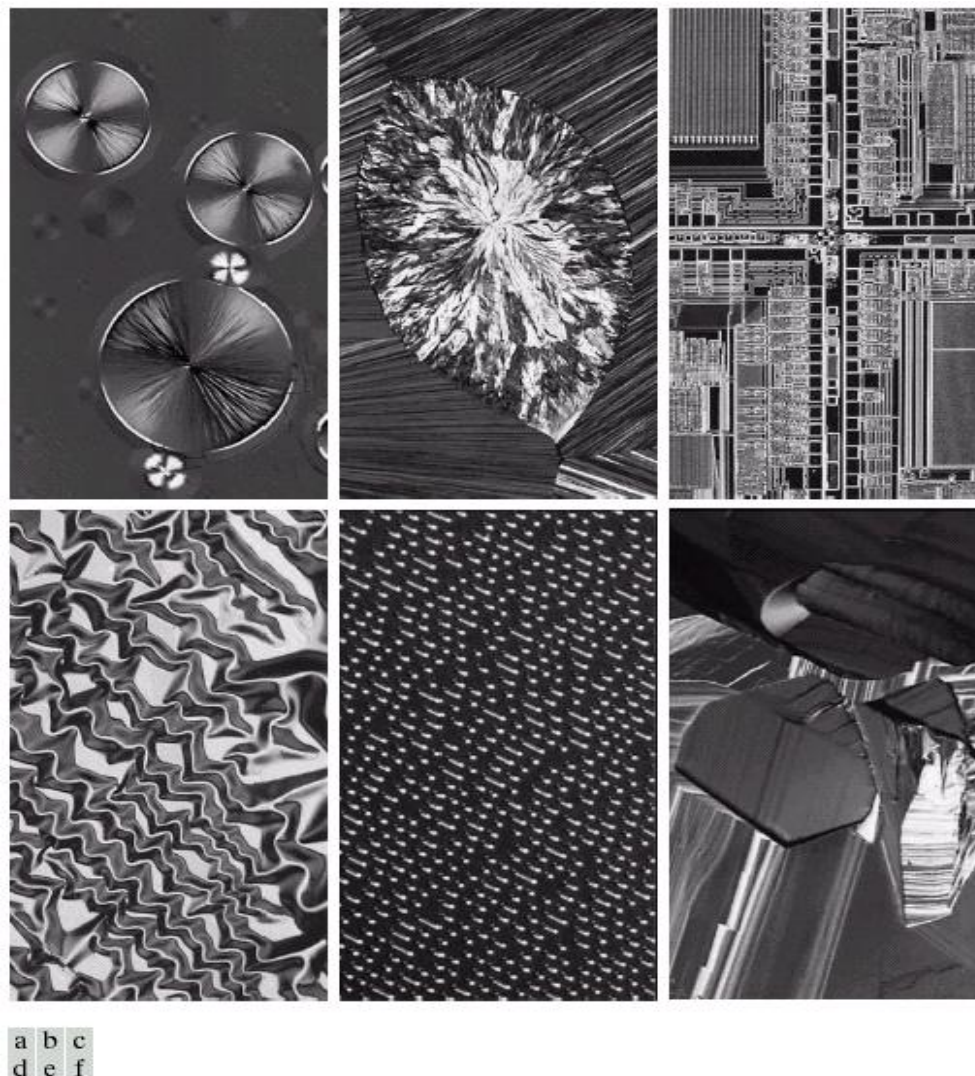


FIGURE 1.9 Examples of light microscopy images. (a) Taxol (anticancer agent), magnified 250 $\times$. (b) Cholesterol—40 $\times$. (c) Microprocessor—60 $\times$. (d) Nickel oxide thin film—600 $\times$. (e) Surface of audio CD—1750 $\times$. (f) Organic superconductor—450 $\times$. (Images courtesy of Dr. Michael W. Davidson, Florida State University.)

TABLE 1.1

Thematic bands
in NASA's
LANDSAT
satellite.

Band No.	Name	Wavelength (μm)	Characteristics and Uses
1	Visible blue	0.45–0.52	Maximum water penetration
2	Visible green	0.52–0.60	Good for measuring plant vigor
3	Visible red	0.63–0.69	Vegetation discrimination
4	Near infrared	0.76–0.90	Biomass and shoreline mapping
5	Middle infrared	1.55–1.75	Moisture content of soil and vegetation
6	Thermal infrared	10.4–12.5	Soil moisture; thermal mapping
7	Middle infrared	2.08–2.35	Mineral mapping

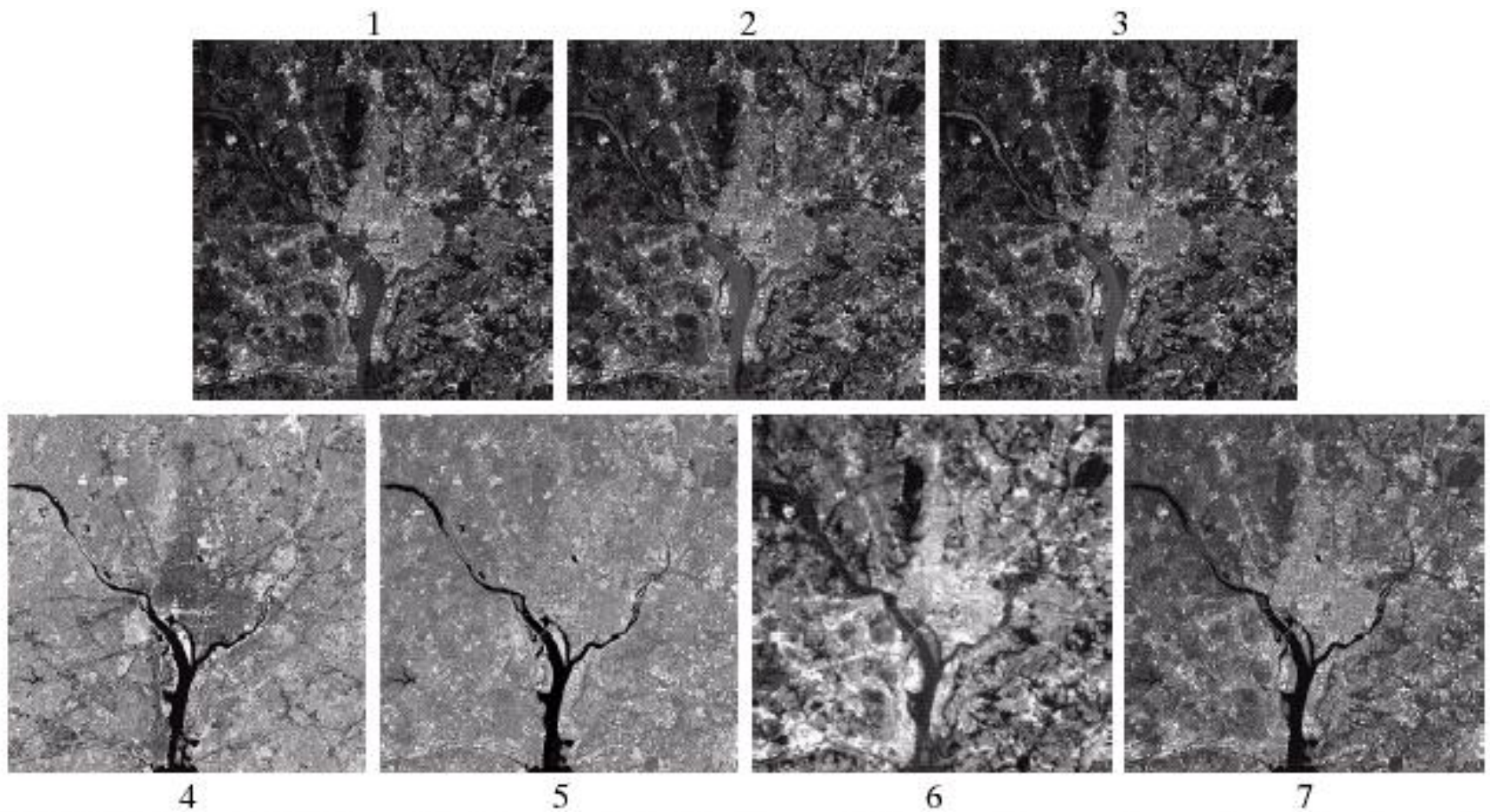
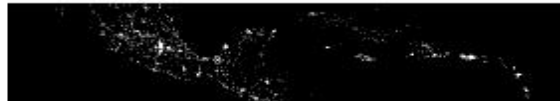


FIGURE 1.10 LANDSAT satellite images of the Washington, D.C. area. The numbers refer to the thematic bands in Table 1.1. (Images courtesy of NASA.)



FIGURE 1.11
Multispectral
image of
Hurricane
Andrew taken by
NOAA GEOS
(Geostationary
Environmental
Operational
Satellite) sensors.
(Courtesy of
NOAA.)

FIGURE 1.12
Infrared satellite
images of the
Americas. The
small gray map
is provided for
reference.
(Courtesy of
NOAA.)



a	b
c	d
e	f

FIGURE 1.14

Some examples of manufactured goods often checked using digital image processing. (a) A circuit board controller.

(b) Packaged pills.

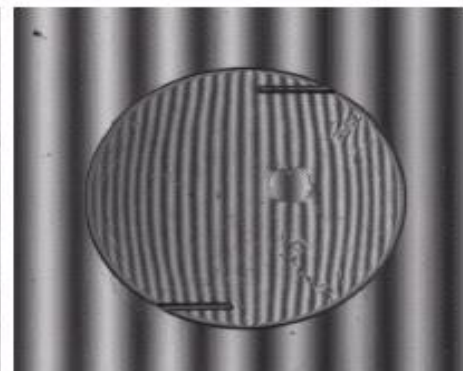
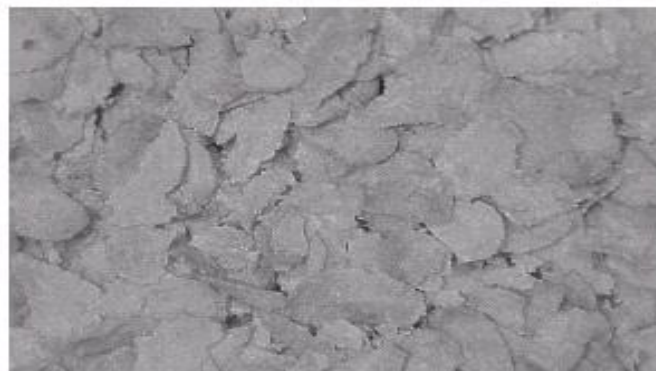
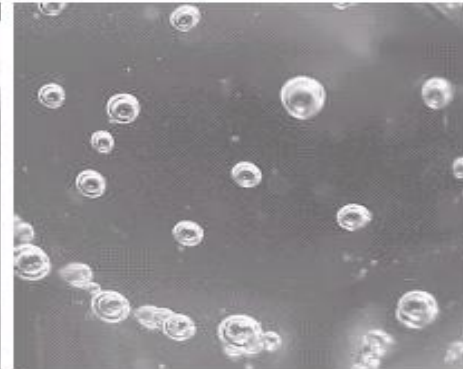
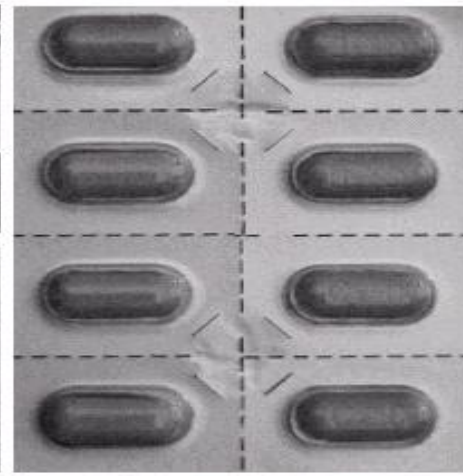
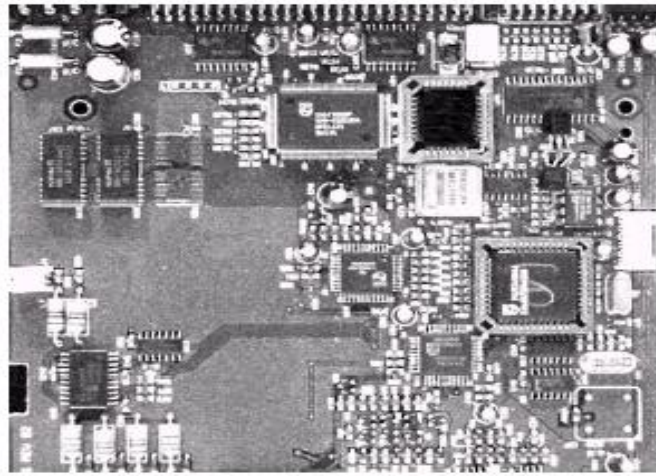
(c) Bottles.

(d) Bubbles in clear-plastic product.

(e) Cereal.

(f) Image of intraocular implant.

(Fig. (f) courtesy of Mr. Pete Sites, Perceptics Corporation.)





a b
c
d

FIGURE 1.15
Some additional examples of imaging in the visual spectrum. (a) Thumb print. (b) Paper currency. (c) and (d). Automated license plate reading. (Figure (a) courtesy of the National Institute of Standards and Technology. Figures (c) and (d) courtesy of Dr. Juan Herrera, Perceptics Corporation.)

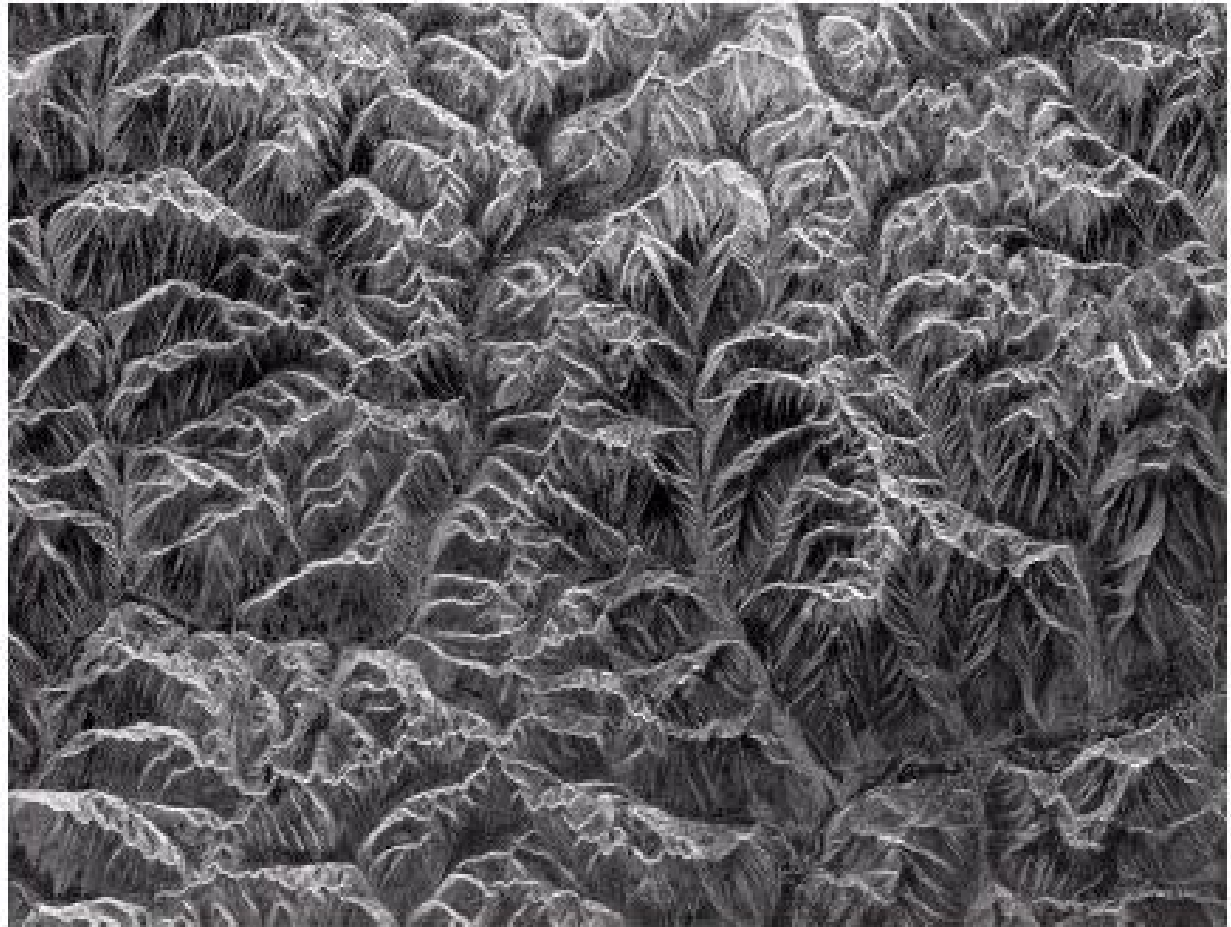


1.3.5 Imaging in the Microwave Band

- **Radar:** imaging radar, regardless of weather or ambient lighting conditions. Fig 1.16

FIGURE 1.16

Spaceborne radar
image of
mountains in
southeast Tibet.
(Courtesy of
NASA.)



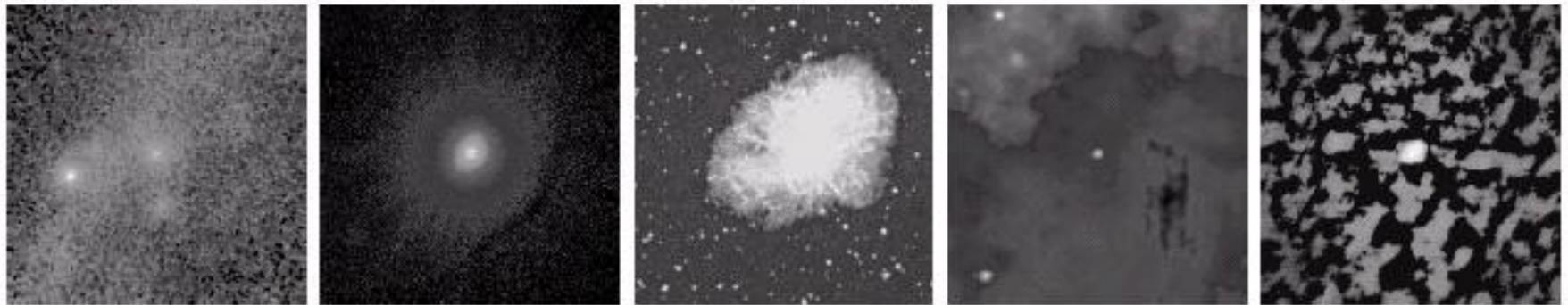
1.3.6 Imaging in the Radio Band

- **Medicine:** Magnetic resonance imaging (MRI). Fig1.17
- **Astronomy:** Crab Pulsar in radio band. Fig1.18.



a b

FIGURE 1.17 MRI images of a human (a) knee, and (b) spine. (Image (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)



Gamma

X-ray

Optical

Infrared

Radio

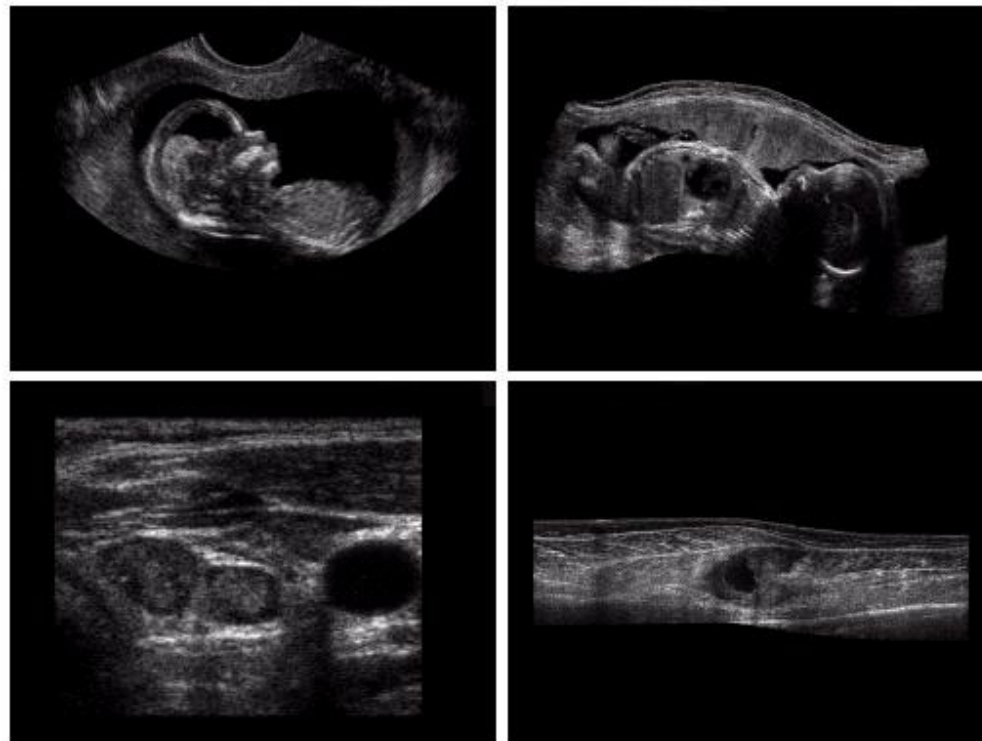
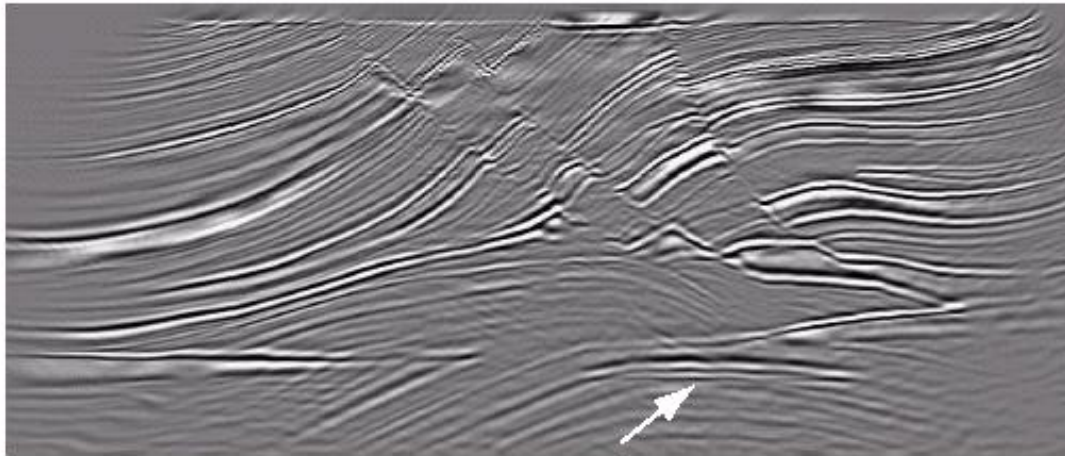
FIGURE 1.18 Images of the Crab Pulsar (in the center of images) covering the electromagnetic spectrum. (Courtesy of NASA.)

1.3.7 Example in Which Other Modalities Are Used

- **Acoustic imaging:** mineral and oil exploration (fig. 1.19) , marine acquisition, obstetrics (fig1.20(a)(b))
- **Electron microscopy:** TEM, SEM, Fig.1.21
- **Synthetic imaging** (computer –generated): Fractals. Fig1.22.

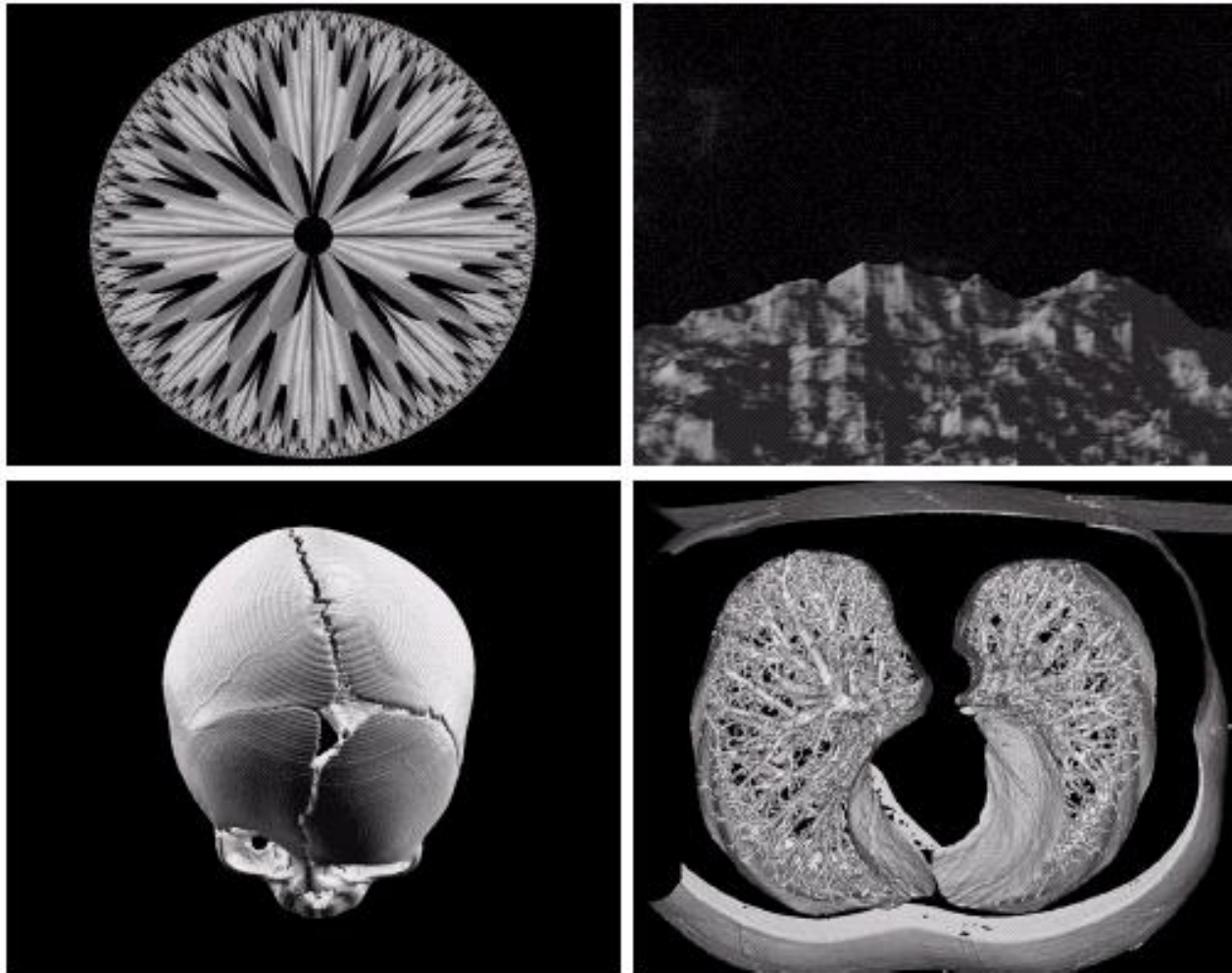
FIGURE 1.19

Cross-sectional image of a seismic model. The arrow points to a hydrocarbon (oil and/or gas) trap. (Courtesy of Dr. Curtis Ober, Sandia National Laboratories.)



a b
c d

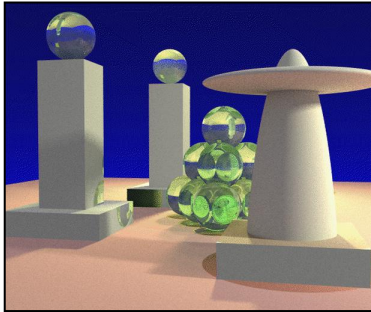
FIGURE 1.20
Examples of ultrasound imaging. (a) Baby. (2) Another view of baby. (c) Thyroids. (d) Muscle layers showing lesion. (Courtesy of Siemens Medical Systems, Inc., Ultrasound Group.)



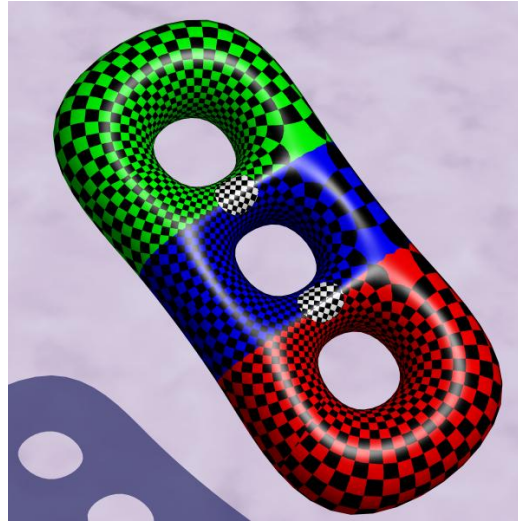
a	b
c	d

FIGURE 1.22

(a) and (b) Fractal images. (c) and (d) Images generated from 3-D computer models of the objects shown. (Figures (a) and (b) courtesy of Ms. Melissa D. Binde, Swarthmore College, (c) and (d) courtesy of NASA.)



Computer graphics



1.4 Others:

Purpose and effects of this course

- learn basic theories and methods of image processing
- easy to use image processing software and equipments
- can develop software and hard ware of image processing
- establish foundation for further research

Appendix:

Components of an Image Processing System

- Sensing and digitizer
- Specialized image processing hardware
- Computer
- Mass storage
- Image displays
- Hardcopy
- Networking
- Software

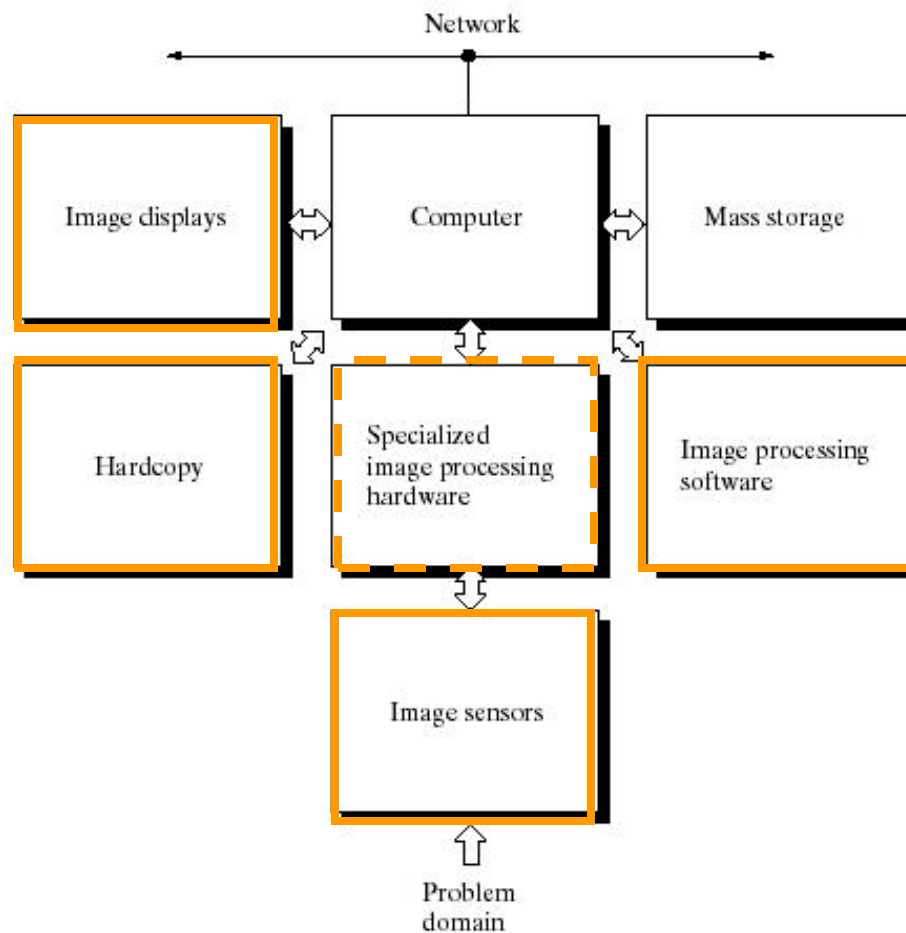


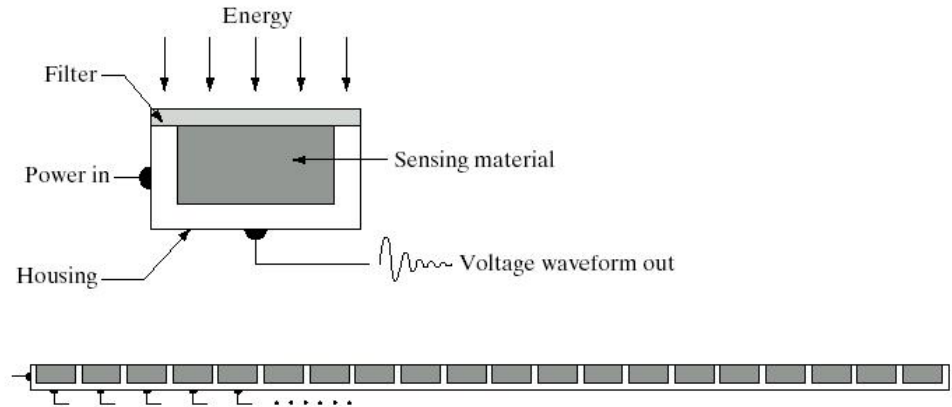
FIGURE 1.24
Components of a
general-purpose
image processing
system.

1. Sensing and digitizer

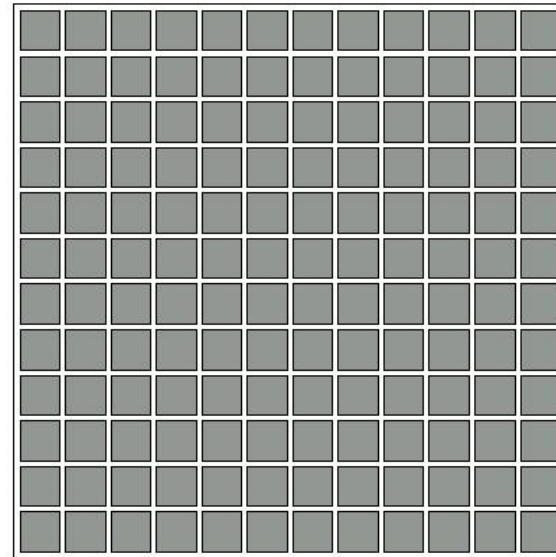
a
b
c

FIGURE 2.12

(a) Single imaging sensor.
(b) Line sensor.
(c) Array sensor.

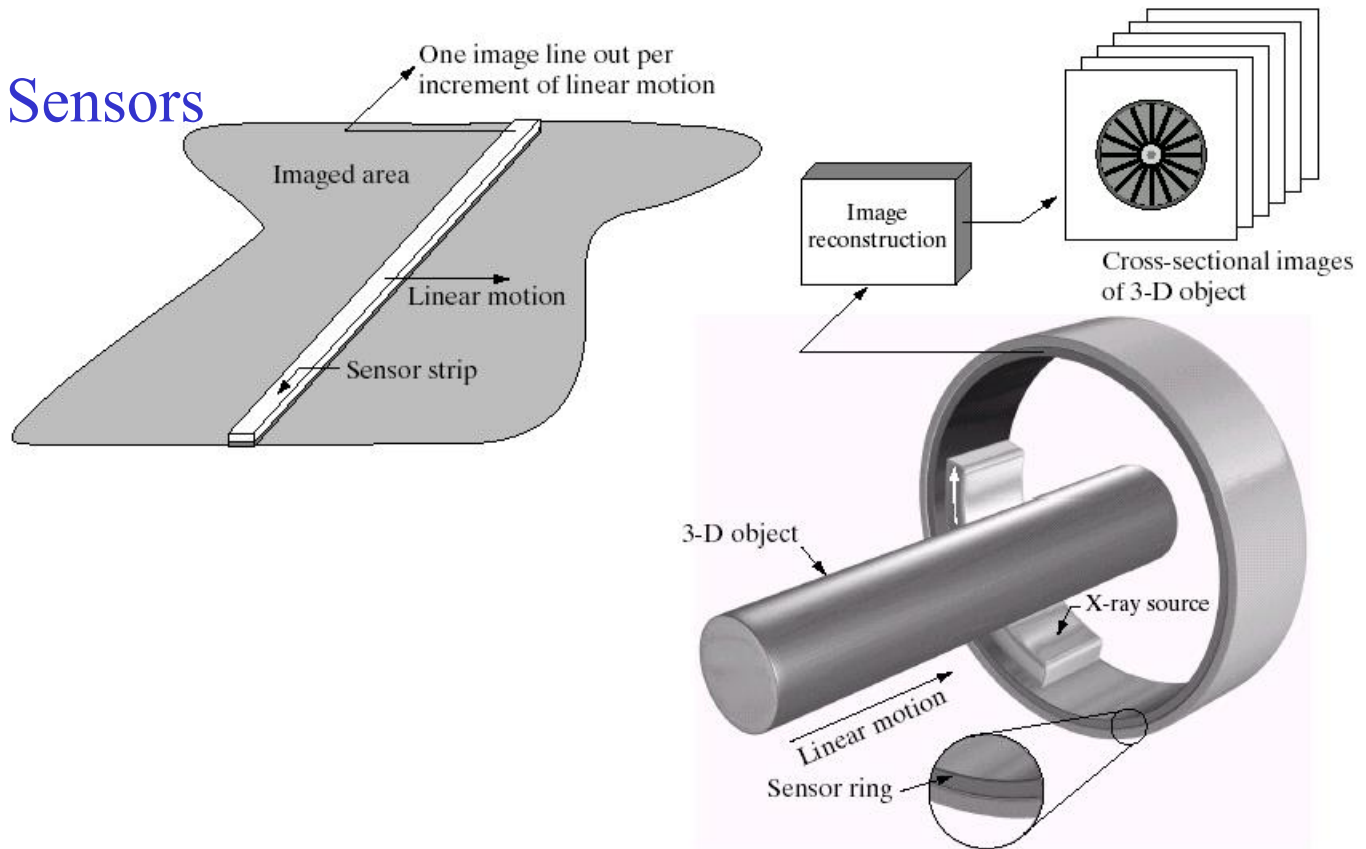


Transform illumination
energy into digital image



1. Sensing and digitizer

Linear Sensors



a b

FIGURE 2.14 (a) Image acquisition using a linear sensor strip. (b) Image acquisition using a circular sensor strip.

1. Sensing and digitizer

Area Sensors

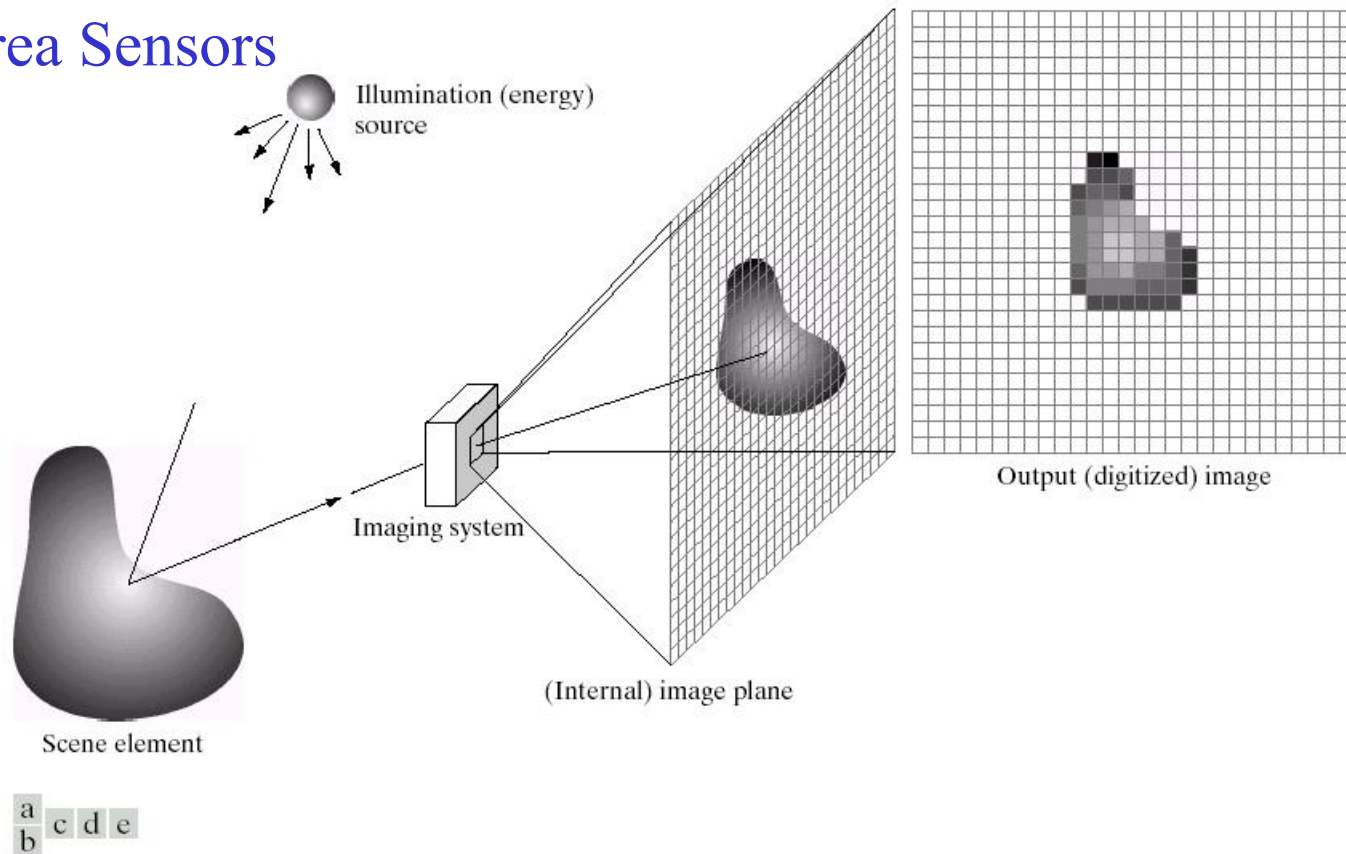
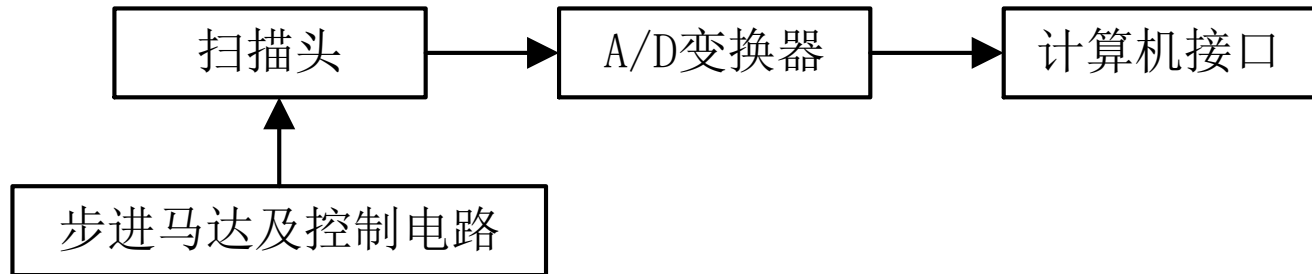


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

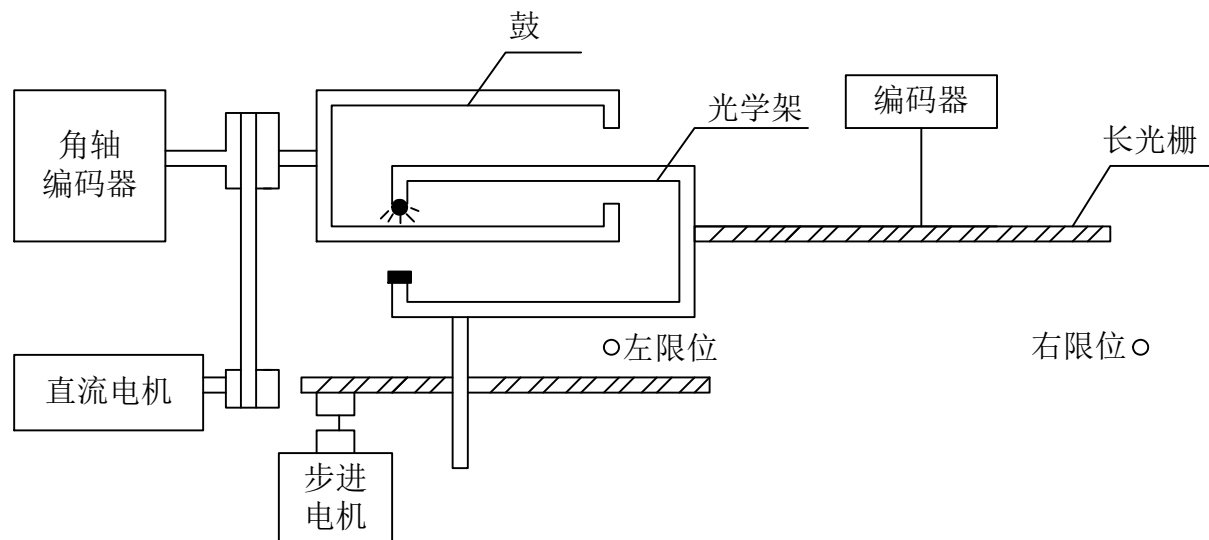
1. Sensing and digitizer

Scanner



扫描仪基本组成框图

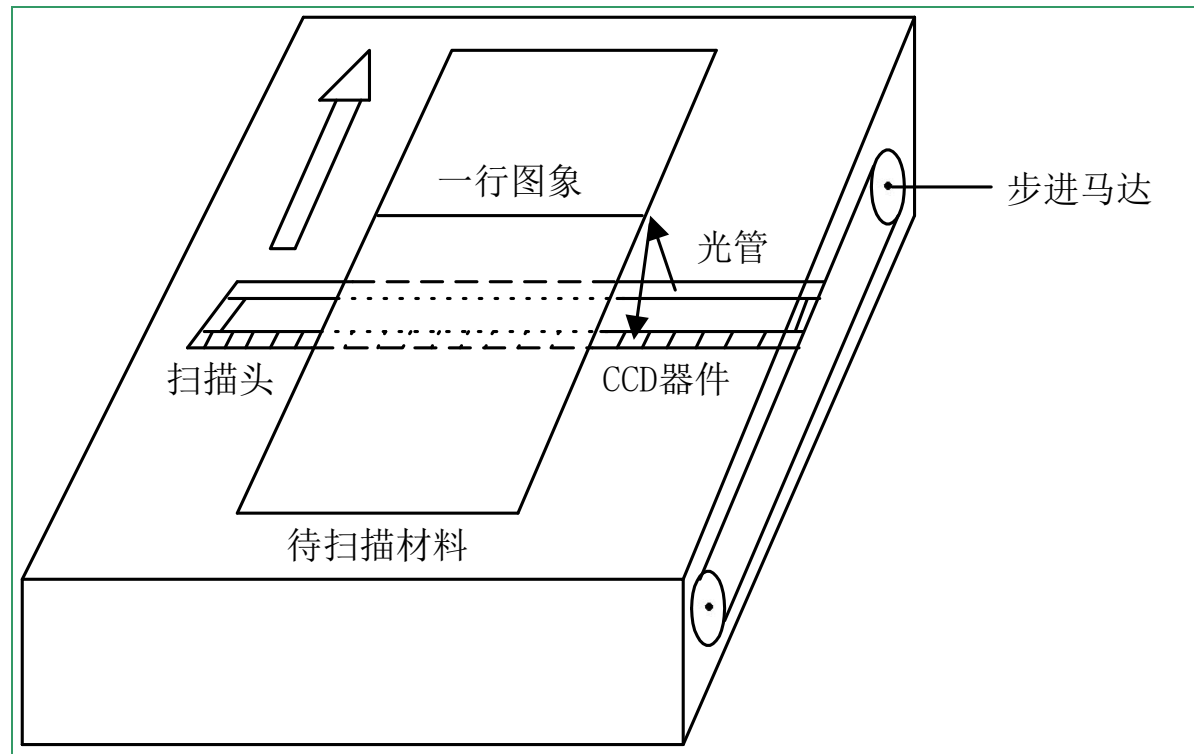
1. Sensing and digitizer



扫描记录仪的结构示意图

1. Sensing and digitizer

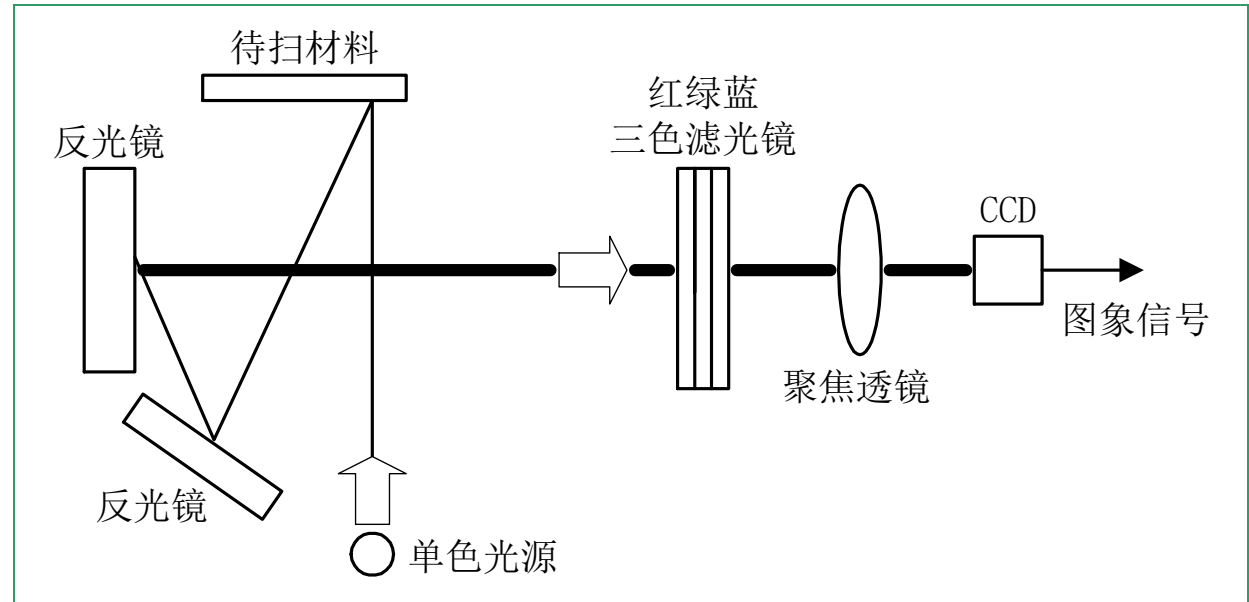
Scanner



扫描仪工作原理示意图

1. Sensing and digitizer

Scanner



彩色图象扫描方式之一

1. Sensing and digitizer

Scanner



Mustek PowerPro

Scanner Type	flat color scanner
Scan Mode: Color Mode Gray Mode Text/Line Art	 24 bits/pixel 8 bits/pixel 1 bits/pixel
Scan Area	216 x 356 mm 8.5" x 14"
Resolution	600dpi x 1200dpi
Scanning Data Buffer	1 MB
Brightness Control	255 adjustable steps
Contrast Control	255 adjustable steps
Light Source	Cold Cathode lamp

1. Sensing and digitizer

Digital Camera



Olympus u300

Price (street)	US\$300
Max resolution	2048 x 1536
Low resolution	1600 x 1200, 1280 x 960, 1024 x 768, 640 x 480
Image ratio w:h	4:3
Effective pixels	3.14 million
Sensor size	1/2.5"
Sensor type	CCD
Colour filter array	G-R-G-B
Max shutter	1/1000 sec



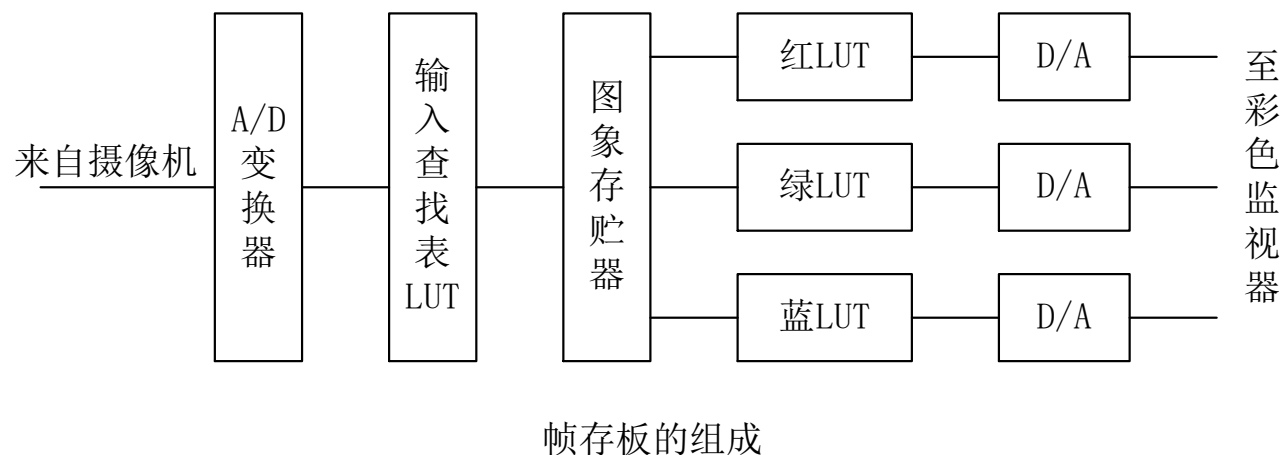
2. Specialized Image Processing Hardware

- PCVISION中的帧存板
- 实时图象处理板MVP-AT
- 实时图象处理卡Cobra/C6

2. Specialized Image Processing Hardware

PCVISION中的帧存板

Look Up Table

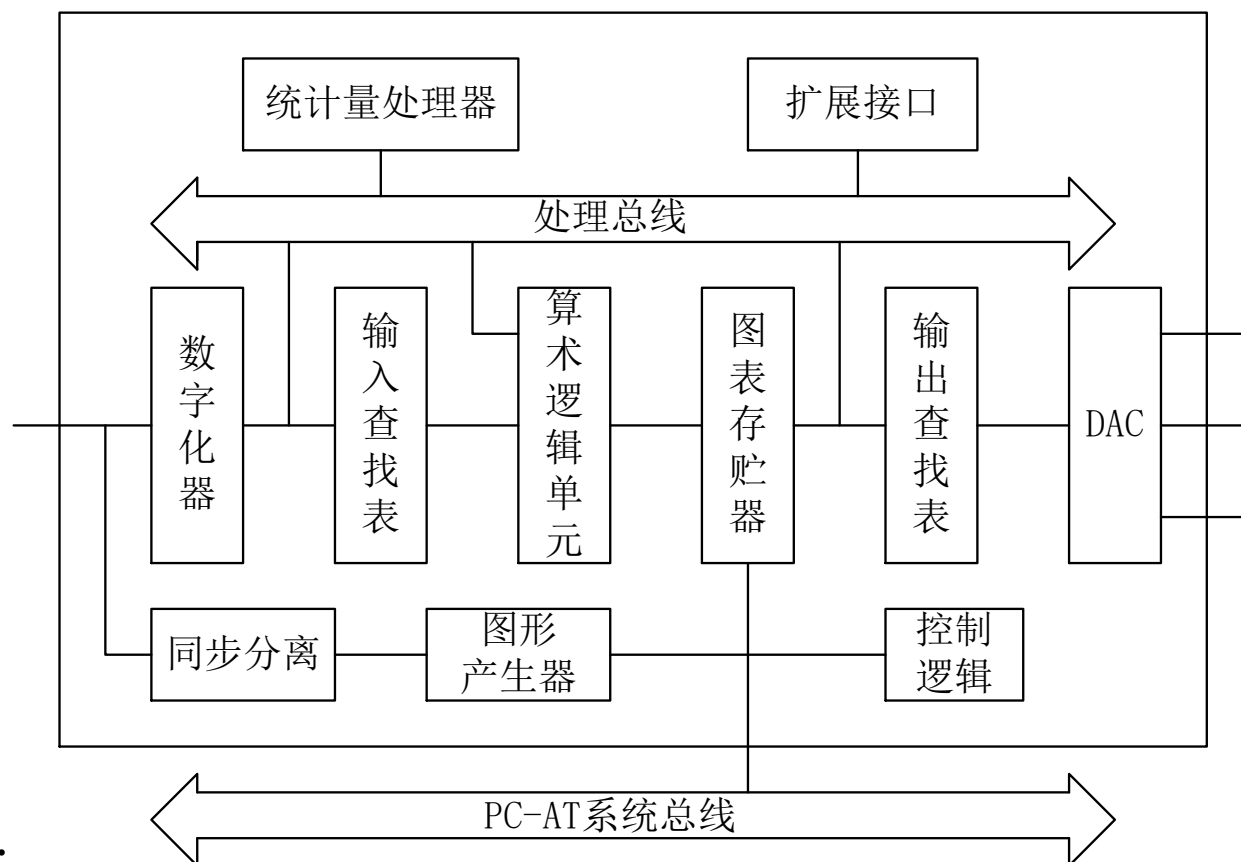


1	1
2	4
3	9
4	16
5	25
6	36
7	49
8	64

Point operation

2. Specialized Image Processing Hardware

实时图象处理板VP-AT



local operation

MVP-AT板方框图
Digital Image Processing
Prof. Zhengkai Liu Dr.Rong Zhang

2. Specialized Image Processing Hardware

实时图象处理卡Cobra/C6

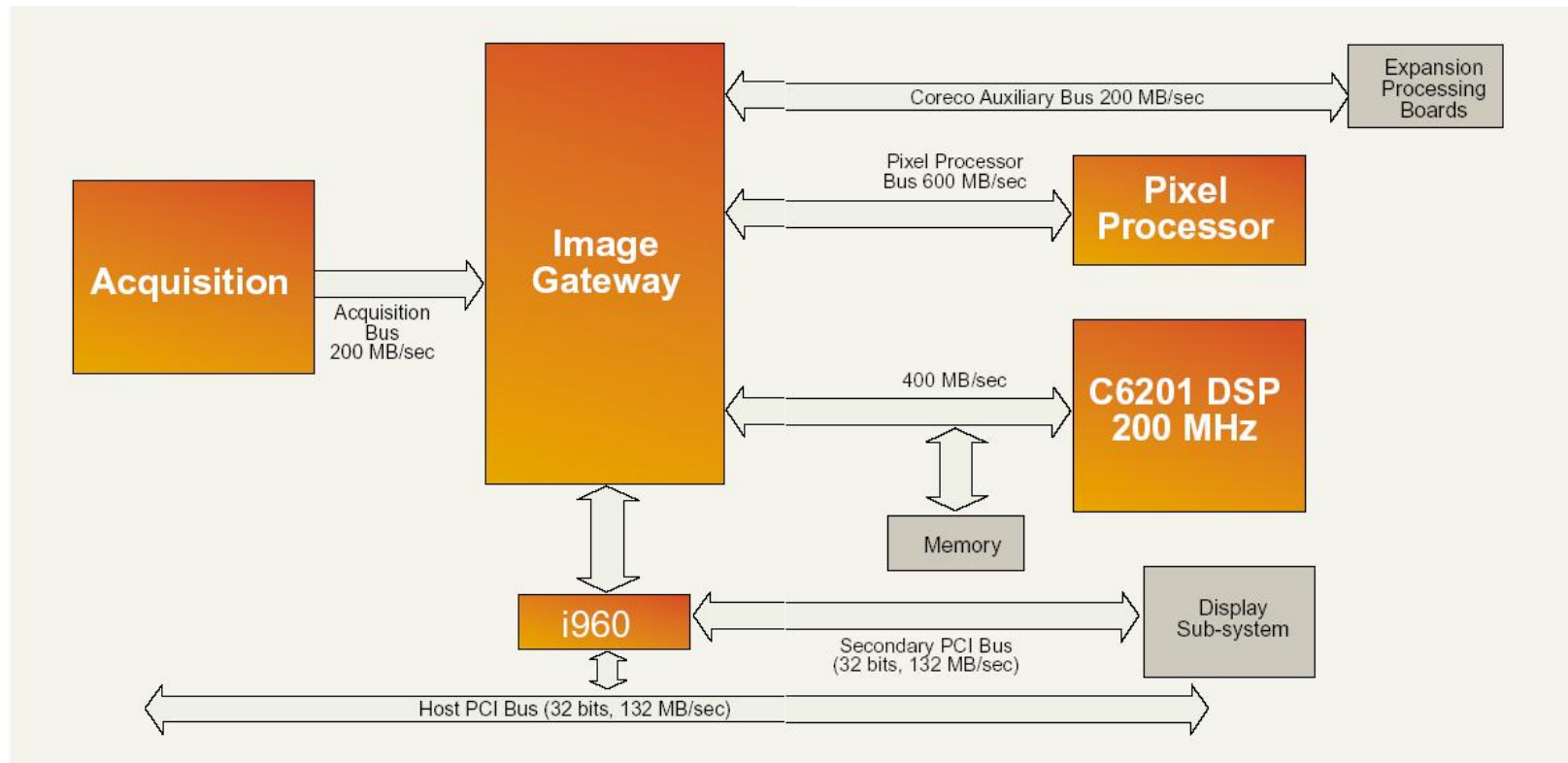


Diagram of the Cobra/C6 architecture
Coreco 公司

2. Specialized Image Processing Hardware

实时图象处理卡Cobra/C6

Pixel Processing

512 x 512 (262144 pixels),

8-bit images.

Category	Function	Speed
Point-to-Point	adding two images	1.3 mSecs
	subtracting two images	1.3 mSecs
	averaging	1.3 mSecs
	unsharp mask	1.3 mSecs
	flat-field correction	1.3 mSecs
Neighborhood operations	convolution (3x3)	1.3 mSecs
	convolution (5x5)	2.6 mSecs
	convolution (7x7)	2.6 mSecs
	convolution (9x9)	5.2 mSecs
	convolution (13x13)	10.5 mSecs
	median Filter	2.6 mSecs
	laplacian/sobel	2.6 mSecs
Statistical operations	histogram	1.3 mSecs
	histogram equalization	2.6 mSecs
Morphology	erosion	1.0 mSecs

2. Specialized Image Processing Hardware

实时图象处理卡Cobra/C6



Key Features:

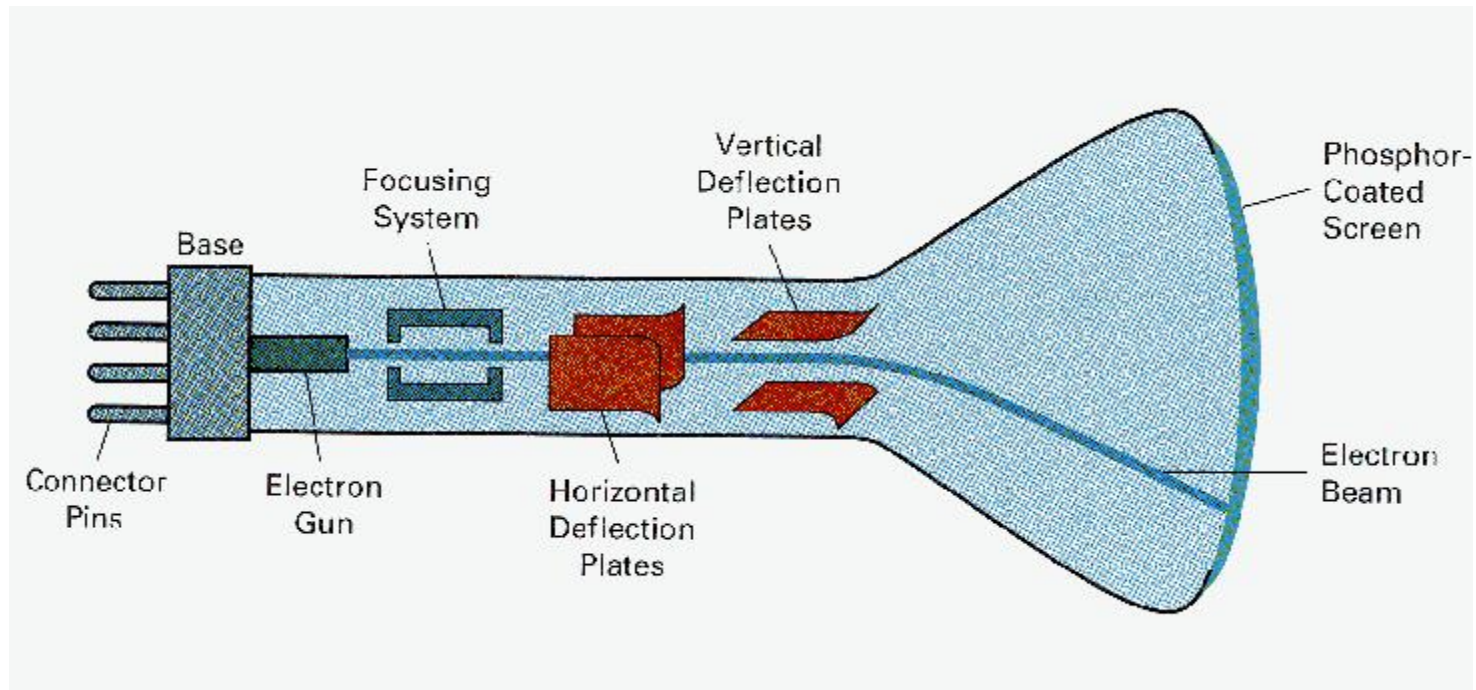
- 200 MHz Operating Frequency
- 64k of high speed on-die cache memory
- 800 Mbytes/sec I/O bandwidth
- 1.6 Gigabyte/sec on-chip transfer rates
- Executes up to eight instructions simultaneously
- Integrated on-chip host port for control
- Six 32/40 bit ALU's
- Two 16 bit single cycle multipliers
- VLIW simplifies parallel programming

3. Image Displays

- CRT(Cathode-Ray Tube)
- Liquid-crystal displays
- Plasma panels

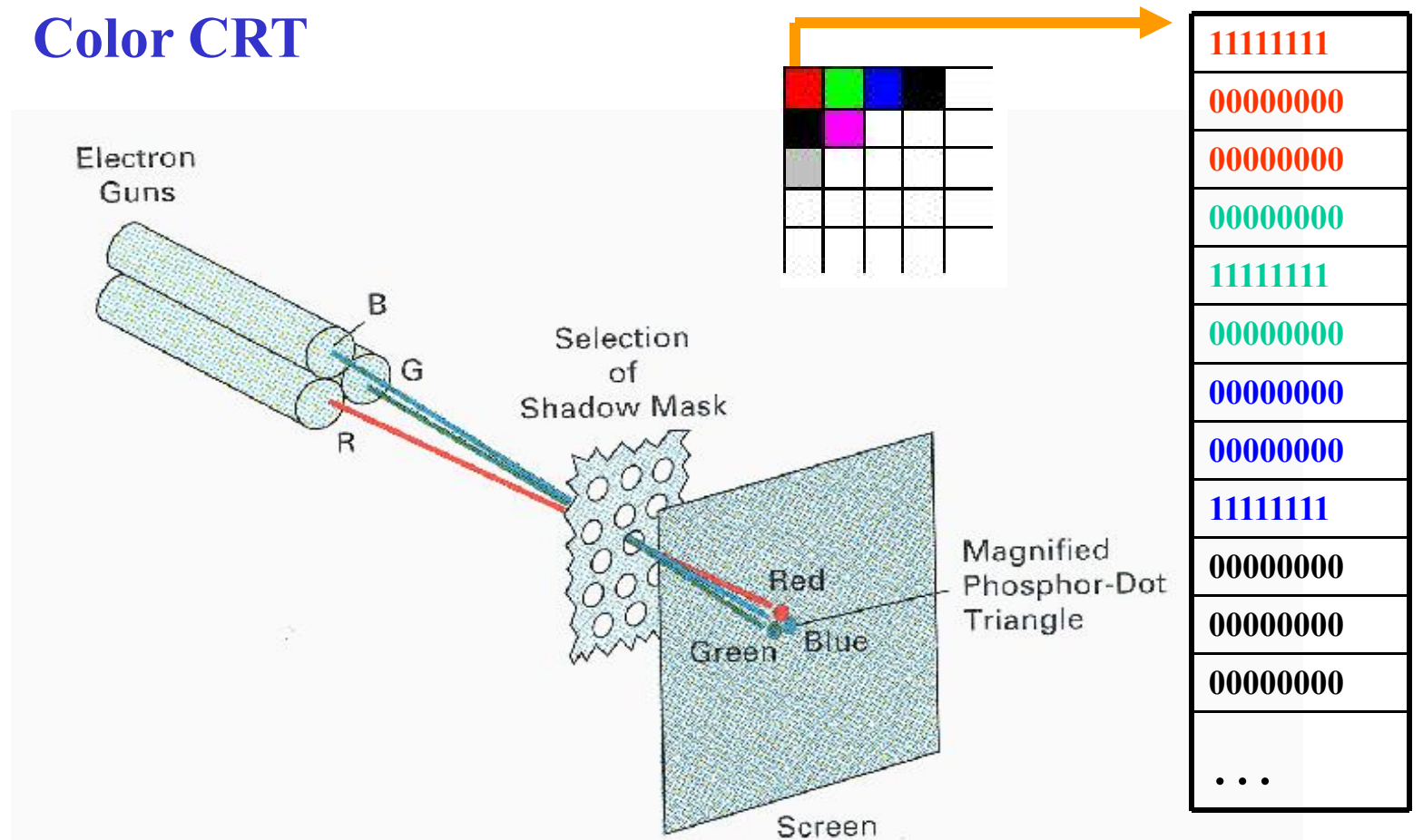
3. Image Displays

CRT



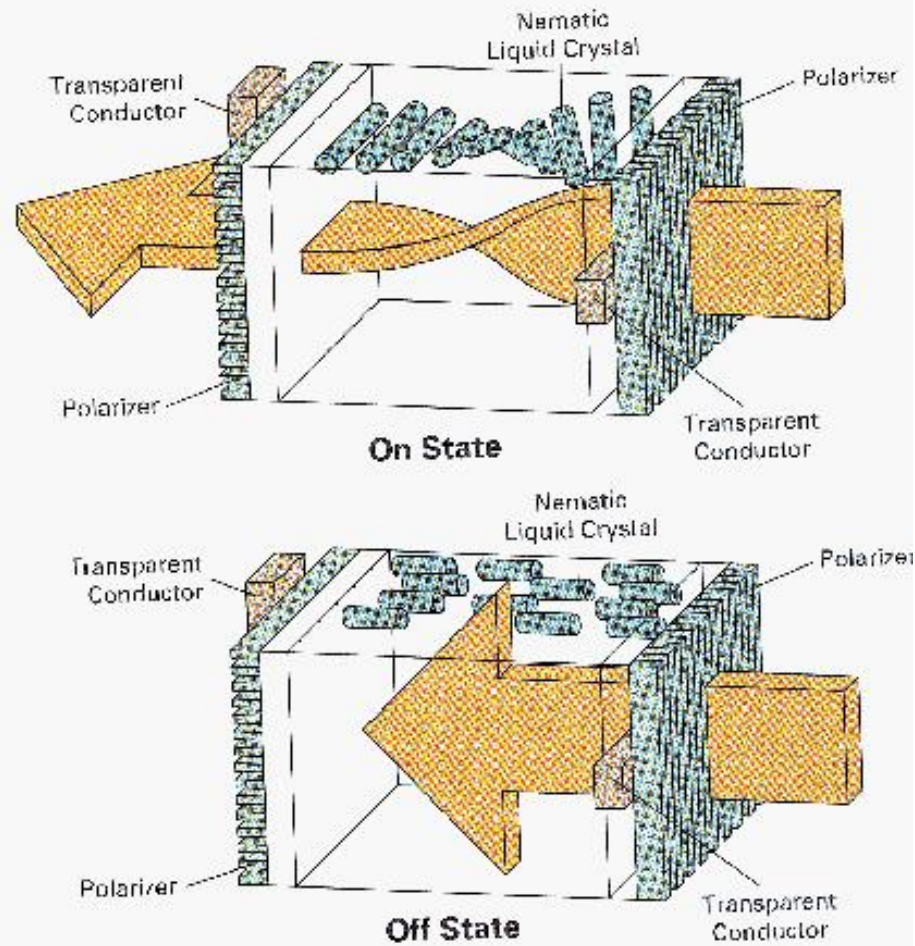
3. Image Displays

Color CRT



3. Image Displays

Liquid Crystal Display

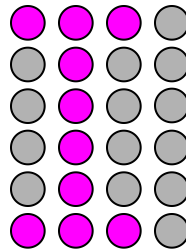


4. Hard Copy

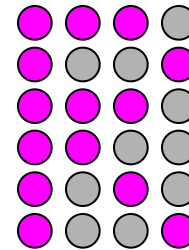
- Dot-matrix Printer
- Laser Printer
- Ink-jet Printer
- Thermal

4. Hard Copy

Dot-matrix Printer

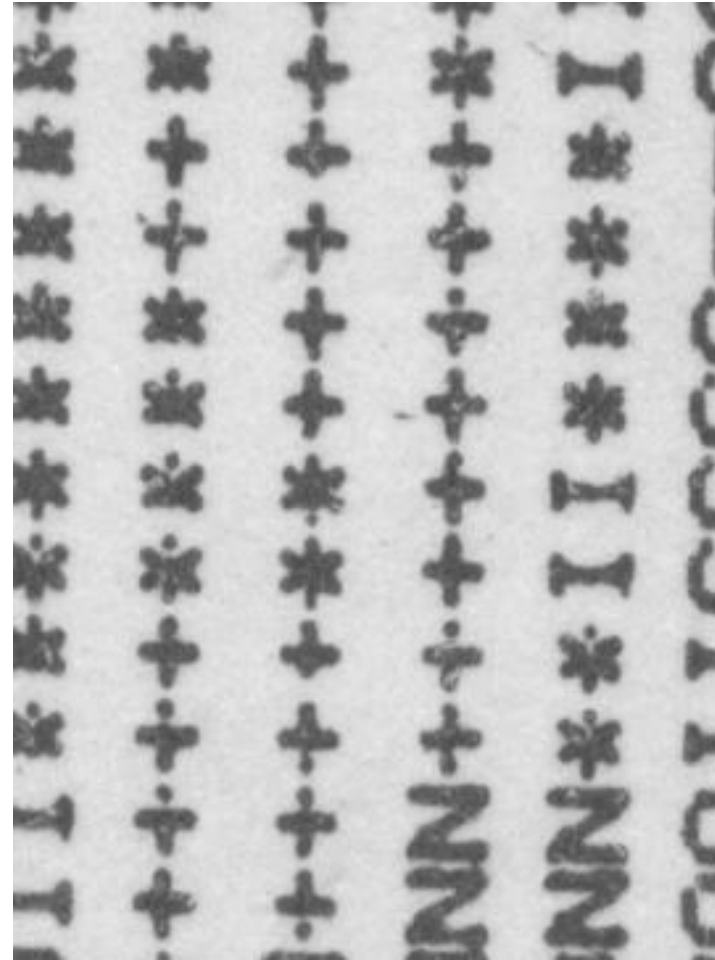


Dot Matrix of “I”



Dot Matrix of “R”

4. Hard Copy



Dither Technique

Digital Image Processing
Prof. Zhengkai Liu Dr.Rong Zhang

4. Hard Copy



Ink-jet Printer

			
1. 插入存储卡。	2. 设定打印张数与用纸尺寸。	3. 按动按键后可自动打印。	

5. Software

- Photoshop
- PhotoImpact
- Corel Custom Photo
- MS PhotoDraw
- Fireworks
- ACDSee

The End